Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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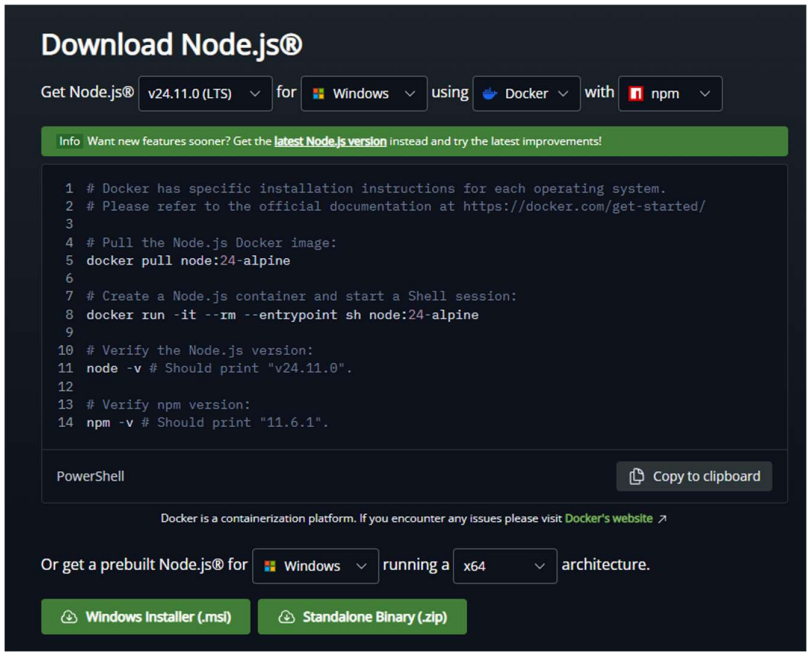
Session	**2025-26 (ODD)**	**Course Name**	**PE-I - Geo-Intelligence for Smart IoT Devices Lab**
Semester	**5**	**Course Code**	**23IOT1523**
Roll No	**64**	**Name of Student**	**Sheikh Wasimuddin**

Practical Number	**09**
Course Outcome	Apply and demonstrate the use of proprietary and open-source GIS tools (e.g., QGIS) for creating, visualizing, and managing spatial datasets.
Aim	Install and implement simple functions of Node-Red.
Problem Definition	The project aims to create a simple Node-RED flow that converts temperature from Celsius to Kelvin and Fahrenheit using an Inject, Function, and Debug node for easy data processing and visualization.
Theory (100 words)	Node-RED is an open-source, flow-based development tool used for visual programming, primarily in Internet of Things (IoT), automation, and data integration projects. It allows users to connect hardware devices, APIs, and online services using a browser-based editor. In this project, Node-RED is utilized to create a simple flow that converts temperature values from Celsius to Kelvin and Fahrenheit. The flow consists of three basic nodes: the Inject node, used to input or trigger a Celsius temperature value; the Function node, which contains JavaScript code to perform the conversion calculations; and the Debug node, which displays the converted results in the debug sidebar for easy visualization. This implementation demonstrates how Node-RED can process and transform data efficiently using minimal coding and a simple drag-and-drop interface, making it ideal for rapid prototyping and educational purposes.

Procedure and Execution (100 Words)	Implementation Steps: 1. Install Nodejs from Browser. 2. After installation of Nodejs, Paste this command in your command line : *npm install -g --unsafe-perm node-red* 3. Once its install, Type node-red in your command line. 4. The node red will start running in you localhost. 5. Add Inject Node: Drag an Inject node and set the payload type to number (e.g., 300 for Kelvin). Rename it as Kelvin Input. 6. Add Function Node: Drag a Function node and insert the conversion code to calculate Celsius and Fahrenheit from the Kelvin input. 7. Add Debug Node: Connect a Debug node to the Function node to display the results on the debug sidebar. 8. Connect Nodes: Link the Inject → Function → Debug nodes in sequence to form the flow. 1. 

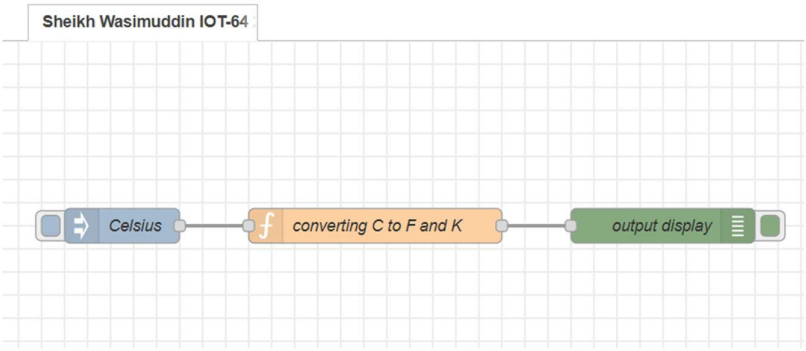
2.

```
Microsoft Windows [Version 10.0.26100.6584]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sheik>npm install node-red

up to date, audited 277 packages in 2s

52 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\sheik>
```

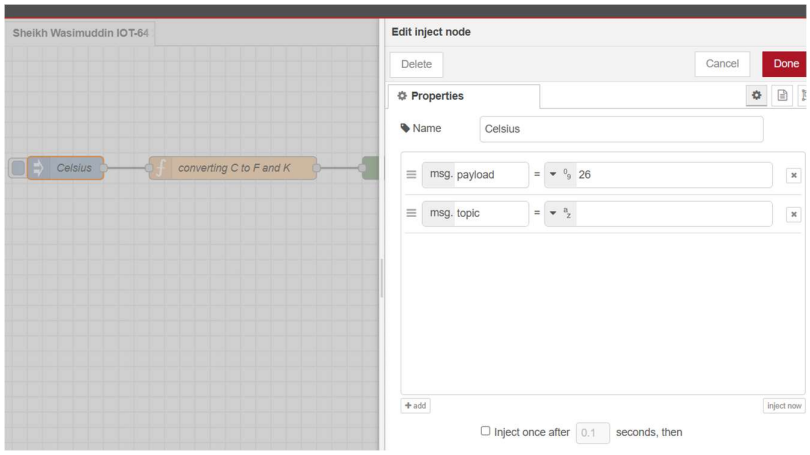
3.

```
C:\Users\sheik>node-red
3 Nov 21:52:59 - [info]

Welcome to Node-RED
===================

3 Nov 21:52:59 - [info] Node-RED version: v4.1.1
3 Nov 21:52:59 - [info] Node.js  version: v22.19.0
3 Nov 21:52:59 - [info] Windows_NT 10.0.26100 x64 LE
3 Nov 21:52:59 - [info] Loading palette nodes
3 Nov 21:53:01 - [info] Settings file  : C:\Users\sheik\.node-red\settings.js
3 Nov 21:53:01 - [info] Context store  : 'default' [module=memory]
3 Nov 21:53:01 - [info] User directory : \Users\sheik\.node-red
3 Nov 21:53:01 - [warn] Projects disabled : editorTheme.projects.enabled=false
3 Nov 21:53:01 - [info] Flows file     : \Users\sheik\.node-red\flows.json
3 Nov 21:53:01 - [warn]

---------------------------------------------------------------------
Your flow credentials file is encrypted using a system-generated key.

If the system-generated key is lost for any reason, your credentials
file will not be recoverable, you will have to delete it and re-enter
your credentials.

You should set your own key using the 'credentialSecret' option in
your settings file. Node-RED will then re-encrypt your credentials
file using your chosen key the next time you deploy a change.
---------------------------------------------------------------------

3 Nov 21:53:01 - [warn] Encrypted credentials not found
3 Nov 21:53:01 - [info] Server now running at http://127.0.0.1:1880/
3 Nov 21:53:01 - [info] Starting flows
3 Nov 21:53:01 - [info] Started flows
3 Nov 21:53:01 - [error] [function:function 1] ReferenceError: msg is not defined
```

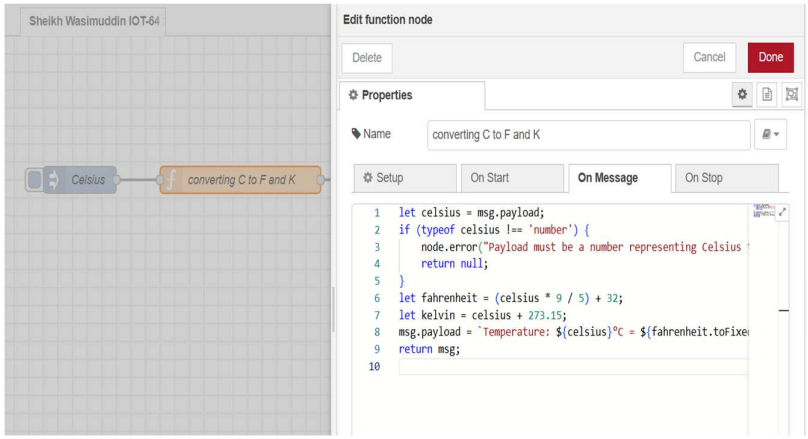
4.



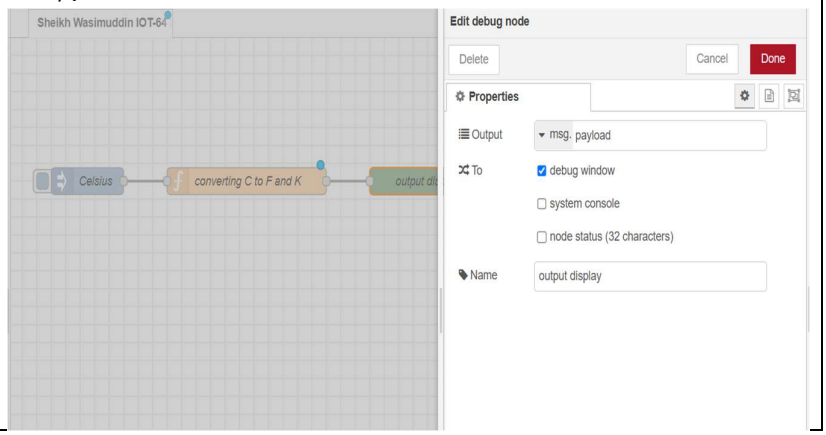
5.

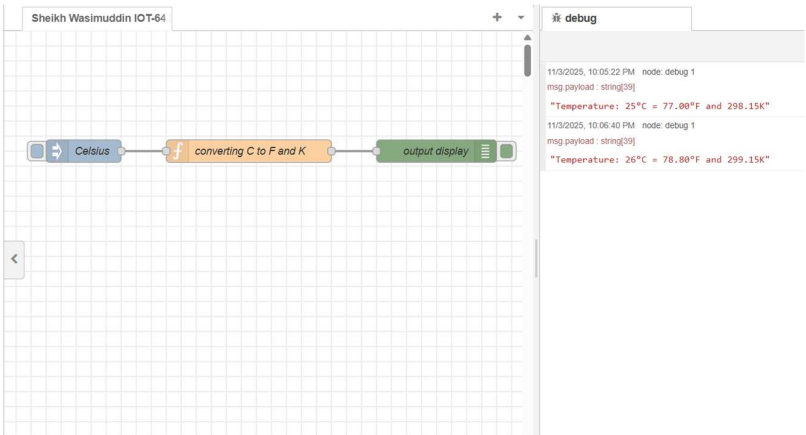

Sheikh Wasimuddin IOT-64
Celsius
converting C to F and K
Edit inject node
Delete
Cancel
Done
Properties
Name
Celsius
msg. payload
26
msg. topic
Inject once after
0.1
seconds, then

6.


Sheikh Wasimuddin IOT-64
Celsius
converting C to F and K
Edit function node
Delete
Cancel
Done
Properties
Name
converting C to F and K
Setup
On Start
On Message
On Stop
let celsius = msg.payload;
if (typeof celsius !== 'number') {
node.error("Payload must be a number representing Celsius
return null;
}
let fahrenheit = (celsius * 9 / 5) + 32;
let kelvin = celsius + 273.15;
msg.payload = `Temperature: ${celsius}°C = ${fahrenheit.toFixe
return msg;

7.


Sheikh Wasimuddin IOT-64
Celsius
converting C to F and K
Edit debug node
Delete
Cancel
Done
Properties
Output
msg. payload
To
debug window
system console
node status (32 characters)
Name
output display

	8. 
Output Analysis	After successfully deploying the Node-RED flow, the Inject node provides a temperature value in Celsius as the input. The Function node processes this input using the embedded JavaScript code to perform real-time conversions into Fahrenheit and Kelvin. The calculated results are formatted into a user-friendly message and displayed in the Debug panel through the Debug node.This output confirms that the flow executes the conversion logic accurately and instantly visualizes the result, demonstrating Node-RED's ability to handle numerical data processing, computation, and output formatting efficiently in a low-code environment.
Conclusion	The implementation of the Node-RED flow for temperature conversion successfully demonstrates how data can be processed and visualized in a simple and efficient manner using flow-based programming. By utilizing the Inject, Function, and Debug nodes, the project converts Celsius input values into Fahrenheit and Kelvin accurately while providing clear and formatted output messages. This exercise highlights the ease of integrating logic and computation within Node-RED, making it an effective tool for IoT applications, real-time data processing, and educational demonstrations of basic data transformation workflows.
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Date	**28/10/2025**